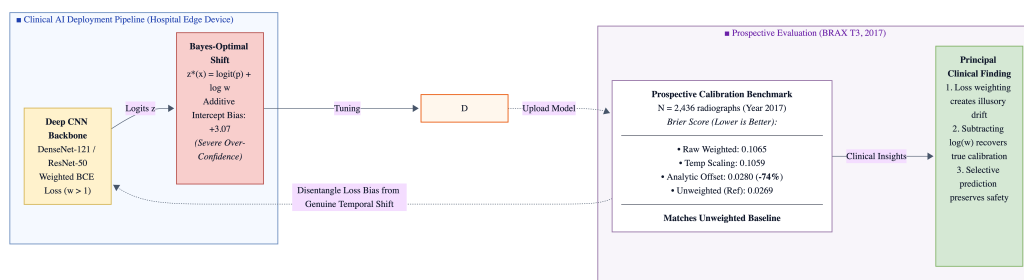


Graphical Abstract

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

Vishesh Panghal



Highlights

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- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 74% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Abstract

Clinical deep learning models frequently exhibit calibration decay under deidentified-date cohort shift. Under extreme class imbalance, training with positive-weighted binary cross-entropy (BCE) is standard practice, but its downstream effect on probability calibration has been widely conflated with intrinsic feature drift. We resolve this conflation through a controlled $2 \times 2 \times 3$ factorial experiment across 12 models (DenseNet-121 and ResNet-50 trained under unweighted and positive-weighted BCE across 3 seeds) on 40,523 chest radiographs from BRAX. Models are trained on an anchor partition and evaluated across later deidentified-date strata (T_1, T_2, T_3) with strict patient isolation. We show that positive loss weighting mathematically induces an additive population logit shift of $\log w$, which temperature scaling structurally cannot correct. In finite networks, empirical logit differences cluster tightly around theoretical $\log w$ (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion). On stratum T_3 , zero-parameter analytic offset ($z - \log w$) reduces total Brier score by 74% ($\Delta \text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]), substantially reducing intercept miscalibration from $\alpha = -2.96$ to -0.45 without retraining or validation data. On an external MIMIC-CXR-JPG cohort (3,403 radiographs, 289 patients), discrimination proved pathology-dependent (Pleural Effusion AUROC 0.688; Cardiomegaly 0.523), with severe baseline calibration shift ($\text{BSS} < 0$). Analytic offset yielded significant Brier reductions for Cardiomegaly and Pneumonia, but worsened

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Edema, delineating boundaries of zero-parameter recalibration under cross-institutional shift. Finally, selective prediction simulations expose operational trade-offs between risk estimation and high-sensitivity screening triage.

Keywords: Chest radiography, Probability calibration, Class imbalance, Distribution shift, Selective prediction, Deep learning, Clinical decision support

1. Introduction

Deep learning models for automated chest radiograph interpretation have achieved high discriminatory performance across common thoracic abnormalities such as pleural effusion, cardiomegaly, and pneumonia [1, 2]. However, the transition from stationary retrospective validation to longitudinal clinical deployment introduces substantial distribution shifts. In operational healthcare environments, patient populations evolve, imaging protocols shift, scanner hardware is replaced, and demographic or seasonal patterns alter condition prevalence over time [3, 4, 5].

While discriminatory ranking (measured via AUROC) often exhibits gradual longitudinal attenuation under temporal shift, decision-critical applications rely fundamentally on *calibrated posterior probabilities*: $P(Y = 1 \mid X = x)$ [6, 7]. When predicted probabilities deviate from true empirical event rates, downstream operational decision cutoffs shift unpredictably, expected utility formulations degrade, and automated decision-support pipelines generate distorted risk rankings.

Recent literature has reported rapid “temporal calibration decay,” asserting that frozen deep neural networks suffer progressive miscalibration over deployment periods [8, 9]. Simultaneously, because biomedical datasets are characterized by extreme class imbalance (pathology prevalence often below 5%), researchers routinely train vision models using positive-weighted binary cross-entropy (BCE) to penalize missed positive cases [1, 10].

In this work, we present a computational investigation showing that these two phenomena are deeply entangled, and that the prevailing attribution of temporal calibration degradation primarily to intrinsic feature drift may be largely explained by uncorrected post-training calibration offsets under label distribution shifts [11, 12]. Under population-risk minimization, when models are trained with positive loss weighting $w > 1$, the Bayes-optimal logit is shifted by an additive constant $+\log w$ [13]. Standard post-hoc

calibration methods—most notably temperature scaling [14]—only rescale the logit variance (z/T) and cannot shift the logit intercept. Consequently, weighted models continuously over-predict positive probabilities, producing severe Brier score and Expected Calibration Error (ECE) degradation that compounds over time as disease prevalence fluctuates.

1.1. Related Work

Probability Calibration in Deep Learning: Modern deep convolutional neural networks frequently exhibit over-confidence due to unconstrained cross-entropy minimization with over-parameterized architectures [14]. Post-hoc recalibration techniques such as temperature scaling, histogram binning, and isotonic regression have been widely evaluated [14, 15]. In medical decision models, calibration accuracy is paramount for valid risk stratification [6, 7]. However, standard methods were primarily derived for multi-class classification under balanced distributions, leaving binary imbalanced settings less formally explored.

Class Imbalance and Loss Function Design: Extreme class imbalance is pervasive in thoracic imaging. Common mitigation strategies include positive loss weighting [10, 1], focal loss [16], and re-sampling. While positive weighting successfully shifts operational sensitivity, King and Zeng [13] proved in classical logistic regression that sample proportion adjustments shift the model intercept by $\log w$. Modern deep learning pipelines routinely apply loss weighting without correcting this induced logit offset prior to post-hoc calibration.

Distribution Shift and Label Shift Adaptation: Temporal dataset shift encompasses changes in feature distributions $P(X)$, conditional distributions $P(Y | X)$, and prior label probabilities $P(Y)$ [3]. Davis et al. [12] demonstrated that calibration drift often precedes discriminatory loss in clinical models. Alexandari et al. [11] established that maximum likelihood with bias-corrected calibration provides robust adaptation under label shift. Decoupling algorithmic training offsets from genuine temporal shift is essential for diagnostic integrity.

Selective Prediction and Deferral Mechanisms: Selective classification allows models to abstain from predicting on ambiguous inputs, routing them to human experts [17, 18]. Fisch et al. [19] highlighted that selective classification performance is fundamentally bounded by probability calibration. Investigating how post-hoc calibration interacts with uncertainty-based triage under temporal shift addresses an open operational question.

67 *1.2. Contributions*

68 To disentangle algorithmic loss bias from distribution shift across cohort
69 strata, this paper provides:

- 70 1. **A Controlled Factorial Matrix:** We execute a $2 \times 2 \times 3$ factorial
71 design across 12 models (DenseNet-121 and ResNet-50 architectures $\times$
72 standard BCE and positive-weighted BCE $\times$ 3 random seeds) trained on
73 25,123 chest radiographs and evaluated on 8,302 held-out radiographs
74 across three sequential deidentified-date strata (T_1, T_2, T_3) from the
75 BRAX dataset [2].
- 76 2. **Theoretical Derivation and Empirical Verification of Logit
77 Shift:** Under population risk minimization, we derive that positive-
78 weighted BCE introduces a constant logit offset $\log w$. In finite neural
79 networks, we verify empirically that observed logit differences $(z_w - z_u)$
80 tightly mirror theoretical $\log w$ across multiple thoracic findings. We
81 show that temperature scaling cannot adjust the logit intercept, leaving
82 the calibration intercept severely biased ($\alpha \approx -2.96$), whereas an
83 exact zero-parameter analytic offset ($z - \log w$) or non-negative Platt
84 scaling restores calibration parity with unweighted models (substantially
85 reducing intercept bias to $\alpha = -0.45$ for DenseNet-121 and reducing
86 total Brier error by 74%).
- 87 3. **A Leak-Free Out-of-Sample Calibration Protocol:** We implement
88 a strict 5-fold patient-clustered cross-fitting protocol on T_1 , strictly
89 preventing in-sample calibration leakage, and report out-of-sample gen-
90 eralization on T_2 and T_3 backed by 2,000-replicate patient-clustered
91 bootstrap confidence intervals directly on individual seeds and the 3-seed
92 ensemble.
- 93 4. **Human-in-the-Loop Selective Prediction Simulation:** We for-
94 mulate a selective triage simulation with explicit, transparent tracking
95 of reader referral workloads, retained sensitivity $(TP/N_{\text{pos, ret}})$, and
96 full-cohort system sensitivity $(1 - FN/N_{\text{pos, total}})$ under matched de-
97 velopment thresholds, demonstrating that calibrated models optimize
98 probability estimation while raw weighted models support recall-oriented
99 screening triage.

100 2. Materials and Methods

101 2.1. Dataset and Cohort Stratification

102 We utilize the Brazilian Chest X-ray (BRAX) v1.1.0 benchmark [2], con-
103 taining 40,523 chest radiographs acquired across multiple inpatient and out-
104 patient emergency facilities at Hospital Israelita Albert Einstein in São Paulo,
105 Brazil. Radiographs are labeled with 14 thoracic findings extracted from
106 Portuguese radiological reports using a validated natural language processing
107 pipeline.

108 In compliance with HIPAA de-identification standards, direct patient iden-
109 tifiers and original timestamps are protected. We extract chronological study
110 dates from the released 8-digit calendar integers (YYYYMMDD), spanning 2008
111 through 2017. As is standard under HIPAA Safe Harbor de-identification,
112 date anonymization may include patient-specific random date shifting that
113 preserves relative within-patient temporal intervals while shifting absolute
114 calendar offsets across patients. Consequently, our prospective partitions
115 T_1 (2015-H2), T_2 (2016), and T_3 (2017) are interpreted as chronologically
116 ordered cohort strata under longitudinal distribution shift rather than syn-
117 chronized real-world calendar years. To maintain strict prospective integrity
118 and preclude longitudinal data leakage:

- 119 1. All studies are partitioned chronologically around a strict model freeze
120 boundary: $T_{\text{freeze}} = 2015-07-01$.
- 121 2. All training studies occur before 2013-09-01 ($N = 25,123$ images from
122 11,265 patients across 14,315 studies).
- 123 3. All validation studies occur between 2013-09-01 and 2015-06-30 ($N =$
124 7,098 images from 3,281 patients across 4,012 studies).
- 125 4. All prospective test studies occur between 2015-07-01 and 2017-12-17
126 ($N = 8,302$ images from 3,807 patients across 4,670 studies), partitioned
127 into three consecutive deployment strata: T_1 (2015-H2, $N = 1,879$), T_2
128 (2016, $N = 3,987$), and T_3 (2017, $N = 2,436$).
- 129 5. Strict patient isolation is enforced: $\text{Patients}_{\text{train}} \cap \text{Patients}_{\text{val}} \cap \text{Patients}_{\text{test}} =$
130 $\emptyset$. A small subset of 89 patients whose visits crossed split boundaries
131 were excluded from validation and testing to guarantee zero within-
132 patient contamination.
- 133 6. For longitudinal paired logit offset and multi-calibrator evaluation, all
134 8,302 prospective radiographs ($N = 1,879$ in T_1 , $N = 3,987$ in T_2 ,
135 and $N = 2,436$ in T_3) are evaluated across both clinical architectures

Table 1: Cohort Demographics, Radiographic Projections, and Target Condition Prevalence across Stratified Cohorts

Cohort Stratum	Images (N)	Patients (N)	Studies (N)	Sex (% M / F)	Age (Mean $\pm$ SD)	Views (% PA / AP / Lat / Unk)	Pleural Effusion N (%)	Cardiomegaly N (%)	Pneumonia N (%)	Edema N (%)
Train (Pre-deployment)	25,123	11,265	14,315	49.3% / 50.7%	46.0 $\pm$ 24.3	34.0% / 13.5% / 0.0% / 27.2%	1,116 (4.44%)	2,398 (9.55%)	456 (1.82%)	23 (0.09%)
Val (Pre-deployment)	7,098	3,281	4,012	49.8% / 50.2%	46.2 $\pm$ 24.1	34.7% / 12.2% / 0.0% / 27.7%	295 (4.16%)	681 (9.59%)	128 (1.80%)	8 (0.11%)
Test T1 (2015)	1,879	913	1,039	47.8% / 52.2%	44.5 $\pm$ 23.9	34.4% / 10.2% / 0.0% / 29.0%	99 (5.27%)	147 (7.82%)	50 (2.66%)	4 (0.21%)
Test T2 (2016)	3,987	1,832	2,249	48.5% / 51.5%	46.1 $\pm$ 23.8	34.6% / 12.0% / 0.0% / 27.6%	177 (4.44%)	422 (10.58%)	79 (1.98%)	13 (0.33%)
Test T3 (2017)	2,436	1,164	1,382	50.9% / 49.1%	46.0 $\pm$ 24.6	34.4% / 12.4% / 0.0% / 28.1%	85 (3.49%)	258 (10.59%)	50 (2.05%)	0 (0.00%)
Test All (Pooled)	8,302	3,807	4,670	49.0% / 51.0%	45.7 $\pm$ 24.1	34.5% / 11.7% / 0.0% / 28.1%	361 (4.35%)	827 (9.96%)	179 (2.16%)	17 (0.20%)

(DenseNet-121 and ResNet-50) with zero prospective exclusions (0% exclusion rate), yielding 16,604 paired model evaluations ($8,302 \times 2$) with complete longitudinal tracking.

Demographic, projection, and disease prevalence characteristics across all cohort strata are detailed in Table 1.

2.2. External Transport Validation Cohort (MIMIC-CXR-JPG)

To evaluate cross-institutional transportability under realistic clinical domain shift, frozen BRAX-trained models were externally evaluated on all eligible frontal radiographs (AP and PA views) from the official patient-partitioned MIMIC-CXR-JPG test partition [20, 21]. The cohort comprises $N = 3,403$ frontal radiographs ($N = 3,041$ diagnostic studies across $N = 289$ unique patients; AP projections: 2,405 [70.67%], PA projections: 998 [29.33%]). In strict accordance with the primary evaluation protocol, ground-truth diagnostic labels were extracted using the official CheXpert labeler under the standard uncertainty-zero (U -zero) policy. Empirical condition prevalence across images was: Pleural Effusion 32.18% ($N = 1,095$ positive radiographs), Cardiomegaly 26.33% ($N = 896$), Edema 21.33% ($N = 726$), and Pneumonia 10.05% ($N = 342$). Because multiple radiographs and studies belong to the same patient, patient-clustered non-parametric bootstrap inference was used throughout external evaluation to account for repeated within-patient observations, avoiding artificial inflation of effective sample size. All 3,403 radiographs were physically verified on disk (100.00% completeness). No retraining or hyperparameter fine-tuning was performed, isolating out-of-distribution transportability from in-domain adaptation.

2.3. Architectures and Training Objectives

We evaluate two standard clinical vision architectures initialized from ImageNet pre-trained weights:

- **DenseNet-121** [22]: Employs dense feature concatenation, standard for thoracic pathology benchmarks [1].

165 • **ResNet-50** [23]: Employs residual skip connections as an independent
 166 structural comparator.

167 Each architecture is trained under two distinct loss objectives:

168 1. **Unweighted Binary Cross-Entropy (Standard BCE):**

$$\mathcal{L}_{\text{std}}(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (1)$$

169 where $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ is the sigmoid probability from raw logit
 170 z .

171 2. **Positive-Weighted Binary Cross-Entropy (Weighted BCE):**

$$\mathcal{L}_{\text{pos}}(y, \hat{p}) = -[w \cdot y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

172 where $w = (N_{\text{total}} - N_{\text{pos}})/N_{\text{pos}}$ represents the inverse positive prevalence
 173 in the training set (e.g., $w = 21.51$ for Pleural Effusion, $w = 9.48$ for
 174 Cardiomegaly, $w = 54.09$ for Pneumonia).

175 Models are trained with the AdamW optimizer (initial learning rate 10^{-4} ,
 176 weight decay 10^{-2}) and a Cosine Annealing learning rate schedule over 15
 177 epochs, with batch size 64 and Automatic Mixed Precision (AMP fp16) on
 178 512×512 pixel input resolution. Checkpoints are selected based on the epoch
 179 achieving the highest validation Pleural Effusion AUROC across the 15 epochs
 180 (occurring between epochs 4 and 15 across runs), without early stopping
 181 loop termination. Each configuration is executed across three random seeds
 182 ($S \in \{42, 123, 2026\}$), producing 12 fully independent models. For unified
 183 system-level benchmarking, a 3-seed probability ensemble is constructed for
 184 each architecture and loss objective by averaging predicted probabilities across
 185 independent training runs:

$$\bar{p}_{\text{raw}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x)), \quad \bar{p}_{\text{corrected}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x) - \log w) \quad (3)$$

186 All post-hoc calibration, paired analytic adjustments, and selective prediction
 187 policies are evaluated directly on both individual model seeds and the unified
 188 probability ensemble, ensuring strict mathematical consistency across all
 189 reported metrics and tables.

190 *2.4. Mathematical Mechanics of Loss Weighting and Calibration*

191 Under positive-weighted binary cross-entropy, the empirical risk minimizer
192 seeks to minimize the expected loss:

$$\begin{aligned}\mathbb{E}_{Y|X}[\mathcal{L}_{\text{pos}}(Y, \hat{p})] &= -w \cdot P(Y = 1 | X) \log \hat{p} \\ &\quad - P(Y = 0 | X) \log(1 - \hat{p})\end{aligned}\tag{4}$$

193 Setting the derivative with respect to logit z to zero:

$$\begin{aligned}\frac{\partial}{\partial z} \mathbb{E}[\mathcal{L}_{\text{pos}}] &= \hat{p} [wP(Y = 1|X) + P(Y = 0|X)] \\ &\quad - wP(Y = 1|X) = 0\end{aligned}\tag{5}$$

194 Solving for the optimal predicted probability $\hat{p}^*$:

$$\hat{p}^* = \frac{w \cdot P(Y = 1 | X)}{w \cdot P(Y = 1 | X) + P(Y = 0 | X)}\tag{6}$$

195 Converting to log-odds:

$$z^*(X) = \log \left(\frac{\hat{p}^*}{1 - \hat{p}^*} \right) = \log \left(\frac{P(Y = 1 | X)}{P(Y = 0 | X)} \right) + \log w\tag{7}$$

196 Equation (7) derives that positive loss weighting shifts the model’s population
197 Bayes-optimal logit by $+\log w$. When $w \gg 1$, the network is mathematically
198 incentivized to output large positive logits, systematically inflating predicted
199 probabilities. We emphasize that Equation (7) describes the population-risk
200 minimizer under infinite capacity; in finite neural networks trained with
201 stochastic gradient descent, empirical logits fluctuate around this theoretical
202 attractor.

203 *2.4.1. Limitation of Temperature Scaling*

204 Temperature scaling [14] adjusts probabilities via a single positive scalar
205 T :

$$p_{\text{temp}} = \sigma \left(\frac{z}{T} \right)\tag{8}$$

206 To formally evaluate calibration structure, we examine the standard logistic
207 calibration regression (Cox calibration) [6, 7]:

$$\text{logit}(P(Y = 1 | X)) = \alpha + \beta \cdot \text{logit}(\hat{p})\tag{9}$$

where $\alpha = 0$ represents perfect calibration-in-the-large (calibration intercept), and $\beta = 1$ denotes ideal calibration slope. Because positive loss weighting induces an additive shift $z \approx z_{\text{true}} + \log w$, the uncorrected logit suffers a massive intercept discrepancy ($\alpha \approx -\log w \ll 0$). Temperature scaling multiplies the logit by $1/T$, adjusting the slope $\beta \rightarrow \beta \cdot T$, but is mathematically incapable of shifting the additive constant α . In low-prevalence clinical domains where negative cases vastly dominate, the negative log-likelihood optimizer is constrained by the conflicting pressures of slope and intercept, typically selecting $T^* \approx 1.0$ or providing limited, architecture-dependent variance shrinkage without eliminating the severe underlying intercept displacement.

2.4.2. Post-Hoc Recalibration Suite

To systematically rectify this mechanism, we evaluate four post-hoc recalibration functions:

1. **Temperature Scaling (T^*):** $p = \sigma(z/T^*)$, optimized via bounded Brent minimization on $[0.01, 50.0]$ with an explicit monotonicity constraint (if $\text{NLL}(T^*) > \text{NLL}(1.0) \implies T^* = 1.0$).
2. **Analytic Offset ($z - \log w$):** Directly subtracts the theoretical training logit shift:

$$p_{\text{analytic}} = \sigma(z - \log w) \quad (10)$$

Requires zero validation data and zero numerical optimization, resetting $\alpha \approx 0$ by construction while preserving model discrimination and slope.

3. **Fitted Offset ($z + a^*$):** Optimizes a single scalar intercept $a^* \in [-10.0, 10.0]$ via univariate Brent minimization on validation log-likelihood, holding the slope fixed at 1.0.
4. **Platt Scaling ($a + bz$):** Logistic regression on logits $p = \sigma(a + bz)$ with a non-negative constraint on slope ($b \geq 0$) to prevent spurious rank reversal.

2.4.3. Algorithmic and Computational Complexity

The evaluated post-hoc calibrators exhibit distinct computational profiles:

- **Temperature Scaling:** Incurs $O(N_{\text{val}})$ scalar optimization over validation logits and $O(1)$ inference per prediction.
- **Analytic Offset:** Incurs strictly $O(1)$ closed-form scalar subtraction ($z - \log w$), requiring zero validation storage, zero parameter fitting, and $O(1)$ inference.

- 241 • **Fitted Offset:** Incurs $O(N_{\text{val}})$ univariate Brent optimization on valida-
242 tion log-likelihood and $O(1)$ inference.
- 243 • **Platt Scaling:** Incurs $O(k \cdot N_{\text{val}})$ iteratively reweighted least squares
244 fitting (where k is the number of optimization steps) and $O(1)$ inference.

245 The analytic offset thus provides an operational advantage for zero-compute
246 edge deployment, eliminating the requirement to retain or access historical
247 validation datasets.

248 2.5. Cross-Fitting and Resampling Evaluation Protocol

249 To ensure zero optimistic in-sample bias:

- 250 • On deployment stratum T_1 (2015-H2), calibrator parameters are esti-
251 mated using **5-fold patient-clustered cross-fitting**. Calibrators are
252 fit on 4 folds and evaluated out-of-fold on the 5th fold, with all studies
253 from any given patient assigned to a single fold.
- 254 • For prospective evaluation on T_2 (2016) and T_3 (2017), the calibrators
255 fit on T_1 are frozen and applied strictly out-of-sample.
- 256 • **Statistical Inference:** All 95% confidence intervals (CIs) are com-
257 puted using $B = 2,000$ patient-clustered bootstrap resamples with
258 replacement. For paired calibrator comparisons (e.g., Analytic Offset
259 vs. Raw), differences are computed on identical bootstrap resamples.

260 2.6. Selective Prediction and Deferral Policy

261 In operational deployment, models need not provide an autonomous pre-
262 diction for every image. An uncertain model can selectively defer ambiguous
263 examinations to expert human readers [17, 18].

264 To ensure fair comparisons across calibrators operating on different proba-
265 bility scales, we determine an operating decision cutoff t^* for each method
266 strictly on development stratum T_1 using Youden’s J index ($J = \text{Sensitivity} +$
267 $\text{Specificity} - 1$), without using prospective evaluation labels. For unweighted
268 models, $t^* \approx 0.014$; for analytic offset calibrated models, $t^* \approx 0.022$; and for
269 raw weighted models, $t^* \approx 0.33$. We define a threshold-relative uncertainty
270 metric based on piecewise distance from the operational cutoff t^* :

$$u(x) = 1 - \frac{|\hat{p}(x) - t^*|}{\mathbb{I}(\hat{p}(x) \geq t^*)(1 - t^*) + \mathbb{I}(\hat{p}(x) < t^*)t^*} \in [0, 1] \quad (11)$$

where $u(x) \rightarrow 1$ indicates maximal ambiguity ($\hat{p}(x) \approx t^*$) and $u(x) \rightarrow 0$ indicates decisive confidence. Crucially, because t^* scales with baseline disease prevalence (e.g., $t^* \approx 0.022$ for analytic calibrated models), predictions below t^* are normalized by t^* . Consequently, low-probability false negatives ($\hat{p} \ll t^*$) yield low uncertainty and are retained in the automated pool rather than referred. At coverage level $C \in [0.5, 1.0]$, the automated system retains the fraction C of cases with lowest uncertainty ($N_{\text{auto}} = \lfloor C \cdot N \rfloor$) and refers the remaining fraction $1 - C$ ($N_{\text{ref}} = N - N_{\text{auto}}$) to radiologist review.

To prevent metric conflation and provide complete arithmetic transparency, we evaluate triage outcomes using distinct, explicit definitions:

- **Retained Positives ($N_{\text{pos, ret}}$):** The number of true positive cases remaining in the automated cohort after ambiguous examinations have been referred to human readers.
- **Automated True Positives (TP_{auto}):** True pathology cases correctly flagged as positive by the autonomous system ($TP_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 1)$).
- **Automated False Negatives (FN_{auto} , Missed Cases):** True pathology cases erroneously classified as negative by the autonomous system ($FN_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 0)$). Note that $TP_{\text{auto}} + FN_{\text{auto}} = N_{\text{pos, ret}}$.
- **Retained Sensitivity:** Diagnostic sensitivity evaluated strictly among the retained automated decisions: $\text{Sens}_{\text{retained}} = \frac{TP_{\text{auto}}}{N_{\text{pos, ret}}} = \frac{TP_{\text{auto}}}{TP_{\text{auto}} + FN_{\text{auto}}}$.
- **Full-Cohort Triage Sensitivity:** System-wide detection sensitivity across all $N_{\text{pos, total}}$ cases in the evaluation stratum (e.g., $N_{\text{pos, total}} = 85$ in T_3), assuming that all examinations deferred to human readers receive expert diagnostic review and are correctly diagnosed: $\text{Sens}_{\text{triage}} = \frac{TP_{\text{auto}} + N_{\text{pos, ref}}}{N_{\text{pos, total}}} = 1 - \frac{FN_{\text{auto}}}{N_{\text{pos, total}}}$.
- **Referral Workload:** The fraction and count of examinations routed to readers ($N_{\text{ref}} = (1 - C) \cdot N$).

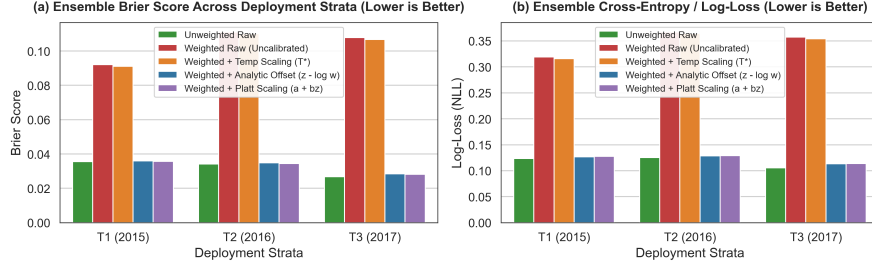


Figure 1: Prospective multi-calibrator evaluation on Pleural Effusion for the DenseNet-121 3-seed ensemble across cohorts (T_1, T_2, T_3), directly visualizing Table 3. (a) Brier score and (b) Log-loss. Temperature scaling fails to reduce probability error in weighted models, whereas Analytic Offset ($z - \log w$) and Platt Scaling eliminate 74% of Brier error, matching unweighted models.

3. Experimental Results

3.1. Factorial Matrix: Discrimination vs. Calibration Trajectories

Table 2 presents the comprehensive evaluation of the $2 \times 2 \times 3$ factorial matrix across DenseNet-121 and ResNet-50 for Pleural Effusion, Cardiomegaly, and Pneumonia across chronological strata T_1, T_2, T_3 .

3.1.1. Discrimination Attenuation

Both architectures exhibit moderate attenuation in AUROC across subsequent deidentified-date cohort strata. For DenseNet-121 on Pleural Effusion, AUROC decreases from 0.9335 ± 0.0052 at T_1 to 0.8731 ± 0.0120 at T_3 ($\Delta\text{AUROC} = -0.0604$). ResNet-50 exhibits an identical downward trajectory ($0.9295 \pm 0.0113 \rightarrow 0.8608 \pm 0.0190$, $\Delta\text{AUROC} = -0.0687$). This confirms that moderate discriminatory attenuation occurs across subsequent cohort strata, regardless of the training loss objective.

3.1.2. The Calibration Divergence and Factorial Interaction

In stark contrast to discrimination, probability calibration diverges completely depending on the training loss function:

- **Unweighted Models:** Brier scores are exceptionally low and actually *improve* across subsequent strata for conditions with falling prevalence. For unweighted DenseNet-121 on Pleural Effusion, the Brier score improves from 0.0369 ± 0.0029 (T_1) to 0.0279 ± 0.0009 (T_3), while Equal-Mass ECE decreases from 0.0232 to 0.0090.

Table 2: Factorial Matrix Evaluation: Longitudinal Trajectories across Architectures, Loss Objectives, and Thoracic Pathologies. Values represent mean $\pm$ standard deviation across three independently trained random seeds ($S \in \{42, 123, 2026\}$).

Architecture	Objective	Target	AUROC T1	AUROC T2	AUROC T3	AUROC Δ	Brier T1	Brier T2	Brier T3	Brier Δ	ECE T1	ECE T2	ECE Δ	LogLoss T1	LogLoss T2	LogLoss Δ
DenseNet-121	Unweighted BCE	Pleural Effusion	0.9335 $\pm$ 0.0052	0.8805 $\pm$ 0.0115	0.8731 $\pm$ 0.0120	-0.0604	0.0309 $\pm$ 0.0029	0.0356 $\pm$ 0.0009	0.0279 $\pm$ 0.0009	-0.0080	0.0232 $\pm$ 0.0078	0.0090 $\pm$ 0.0022	-0.0143	0.1318 $\pm$ 0.0076	0.1114 $\pm$ 0.0027	-0.0204
DenseNet-121	Unweighted BCE	Cardiomegaly	0.9075 $\pm$ 0.0116	0.8965 $\pm$ 0.0140	0.9017 $\pm$ 0.0170	-0.0057	0.0532 $\pm$ 0.0012	0.0677 $\pm$ 0.0021	0.0677 $\pm$ 0.0028	+0.0145	0.0140 $\pm$ 0.0009	0.0210 $\pm$ 0.0074	+0.0071	0.1798 $\pm$ 0.0072	0.2231 $\pm$ 0.0102	+0.0433
DenseNet-121	Unweighted BCE	Pneumonia	0.8473 $\pm$ 0.0129	0.8818 $\pm$ 0.0106	0.8647 $\pm$ 0.0172	+0.0174	0.0235 $\pm$ 0.0008	0.0182 $\pm$ 0.0003	0.0157 $\pm$ 0.0006	-0.0048	0.0142 $\pm$ 0.0011	0.0095 $\pm$ 0.0016	-0.0046	0.0998 $\pm$ 0.0035	0.0888 $\pm$ 0.0039	-0.0107
DenseNet-121	Unweighted BCE	Edema*	0.5402 $\pm$ 0.0467	0.8640 $\pm$ 0.0686	-	-	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0029 $\pm$ 0.0013	0.0029 $\pm$ 0.0029	-0.0000	0.0053 $\pm$ 0.0008	0.0029 $\pm$ 0.0029	-0.0134
DenseNet-121	Weighted BCE	Pleural Effusion	0.9661 $\pm$ 0.0537	0.8676 $\pm$ 0.0457	0.8391 $\pm$ 0.0657	-0.0670	0.1081 $\pm$ 0.0394	0.1287 $\pm$ 0.0432	0.1254 $\pm$ 0.0456	+0.0173	0.1964 $\pm$ 0.0702	0.2227 $\pm$ 0.0765	+0.0264	0.3646 $\pm$ 0.1179	0.4100 $\pm$ 0.1279	+0.0454
DenseNet-121	Weighted BCE	Cardiomegaly	0.8739 $\pm$ 0.0540	0.8589 $\pm$ 0.0600	0.8575 $\pm$ 0.0609	-0.0164	0.1262 $\pm$ 0.0247	0.1358 $\pm$ 0.0330	0.1300 $\pm$ 0.0306	+0.0088	0.2031 $\pm$ 0.0507	0.1969 $\pm$ 0.0486	-0.0062	0.3862 $\pm$ 0.0807	0.4241 $\pm$ 0.0906	+0.0379
DenseNet-121	Weighted BCE	Pneumonia	0.8181 $\pm$ 0.0206	0.8494 $\pm$ 0.0313	0.8424 $\pm$ 0.0496	+0.0243	0.1313 $\pm$ 0.0533	0.1308 $\pm$ 0.0555	0.1298 $\pm$ 0.0513	-0.0015	0.2393 $\pm$ 0.0894	0.2425 $\pm$ 0.0883	+0.0032	0.4226 $\pm$ 0.1511	0.4166 $\pm$ 0.1471	-0.0060
DenseNet-121	Weighted BCE	Edema*	0.6122 $\pm$ 0.0268	0.8753 $\pm$ 0.0129	-	-	0.0614 $\pm$ 0.0637	0.0672 $\pm$ 0.0763	0.0636 $\pm$ 0.0734	-0.0022	0.1191 $\pm$ 0.1228	0.1274 $\pm$ 0.1327	+0.0082	0.2095 $\pm$ 0.2021	0.2093 $\pm$ 0.2288	-0.0002
ResNet-50	Unweighted BCE	Pleural Effusion	0.9295 $\pm$ 0.0113	0.8752 $\pm$ 0.0293	0.8608 $\pm$ 0.0190	-0.0687	0.0372 $\pm$ 0.0022	0.0364 $\pm$ 0.0013	0.0281 $\pm$ 0.0010	-0.0088	0.0162 $\pm$ 0.0050	0.0132 $\pm$ 0.0030	-0.0029	0.1291 $\pm$ 0.0099	0.1145 $\pm$ 0.0067	-0.0146
ResNet-50	Unweighted BCE	Cardiomegaly	0.8898 $\pm$ 0.0177	0.8756 $\pm$ 0.0317	0.8785 $\pm$ 0.0196	-0.0112	0.0572 $\pm$ 0.0037	0.0709 $\pm$ 0.0059	0.0712 $\pm$ 0.0051	+0.0040	0.0233 $\pm$ 0.0190	0.0175 $\pm$ 0.0081	-0.0058	0.1954 $\pm$ 0.0193	0.2379 $\pm$ 0.0174	+0.0425
ResNet-50	Unweighted BCE	Pneumonia	0.8432 $\pm$ 0.0178	0.8659 $\pm$ 0.0130	0.8785 $\pm$ 0.0199	+0.0366	0.0242 $\pm$ 0.0007	0.0187 $\pm$ 0.0003	0.0191 $\pm$ 0.0003	0.0051	0.0128 $\pm$ 0.0050	0.0106 $\pm$ 0.0015	-0.0023	0.1024 $\pm$ 0.0059	0.0889 $\pm$ 0.0035	-0.0135
ResNet-50	Unweighted BCE	Edema*	0.5564 $\pm$ 0.0368	0.8566 $\pm$ 0.0516	-	-	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0023 $\pm$ 0.0005	0.0017 $\pm$ 0.0009	-0.0006	0.0175 $\pm$ 0.0010	0.0117 $\pm$ 0.0009	-0.0158
ResNet-50	Weighted BCE	Pleural Effusion	0.8789 $\pm$ 0.0508	0.8467 $\pm$ 0.0458	0.8191 $\pm$ 0.0496	-0.0598	0.1256 $\pm$ 0.0325	0.1417 $\pm$ 0.0277	0.1392 $\pm$ 0.0310	+0.0137	0.2481 $\pm$ 0.0733	0.2754 $\pm$ 0.0709	+0.0273	0.4003 $\pm$ 0.0895	0.4365 $\pm$ 0.0750	+0.0361
ResNet-50	Weighted BCE	Cardiomegaly	0.8208 $\pm$ 0.0550	0.8046 $\pm$ 0.0701	0.7990 $\pm$ 0.0788	-0.0308	0.1415 $\pm$ 0.0124	0.1526 $\pm$ 0.0083	0.1522 $\pm$ 0.0085	+0.0107	0.2459 $\pm$ 0.0348	0.2338 $\pm$ 0.0256	-0.0121	0.4427 $\pm$ 0.0380	0.4678 $\pm$ 0.0305	+0.0251
ResNet-50	Weighted BCE	Pneumonia	0.8349 $\pm$ 0.0199	0.8359 $\pm$ 0.0345	0.8233 $\pm$ 0.0533	-0.0116	0.1667 $\pm$ 0.0584	0.1681 $\pm$ 0.0583	0.1691 $\pm$ 0.0613	+0.0024	0.3189 $\pm$ 0.1175	0.3253 $\pm$ 0.1208	+0.0064	0.5056 $\pm$ 0.1443	0.5096 $\pm$ 0.1527	+0.0040
ResNet-50	Weighted BCE	Edema*	0.5490 $\pm$ 0.0852	0.8310 $\pm$ 0.0437	-	-	0.0903 $\pm$ 0.0461	0.0749 $\pm$ 0.0509	0.0746 $\pm$ 0.0494	+0.0052	0.1545 $\pm$ 0.1121	0.1670 $\pm$ 0.1160	+0.0126	0.2193 $\pm$ 0.1374	0.2295 $\pm$ 0.1469	+0.0102

- **Positive-Weighted Models:** Brier scores are dramatically elevated and appear to degrade across subsequent cohort strata ($0.1081 \pm 0.0394 \rightarrow 0.1254 \pm 0.0456$ for DenseNet-121; $0.1256 \pm 0.0325 \rightarrow 0.1392 \pm 0.0310$ for ResNet-50). ECE exceeds 22% across all prospective strata.

To formally examine the interaction between loss weighting and temporal cohort (LossWeighting $\times$ TemporalCohort), we evaluate the excess calibration error $\Delta\text{Brier} = \text{Brier}_{\text{weighted}} - \text{Brier}_{\text{unweighted}}$ across sequential strata:

- **Pleural Effusion (Prevalence Declines: 5.27% $\rightarrow$ 4.44% $\rightarrow$ 3.49%):**
The excess calibration penalty of positive weighting expands monotonically across cohorts, from $\Delta\text{Brier} = +0.0712 \pm 0.0413$ in T_1 to $+0.0932 \pm 0.0428$ in T_2 and $+0.0975 \pm 0.0454$ in T_3 .
- **Cardiomegaly (Prevalence Increases: 7.82% $\rightarrow$ 10.58% $\rightarrow$ 10.59%):**
The excess calibration penalty remains stable or slightly attenuates: $\Delta\text{Brier} = +0.0729 \pm 0.0255$ in T_1 , $+0.0681 \pm 0.0329$ in T_2 , and $+0.0683 \pm 0.0320$ in T_3 .
- **Pneumonia (Prevalence Steady Low: 2.66% $\rightarrow$ 1.98% $\rightarrow$ 2.05%):**
The penalty remains consistently elevated: $\Delta\text{Brier} = +0.1079 \pm 0.0526$ in T_1 , $+0.1126 \pm 0.0552$ in T_2 , and $+0.1112 \pm 0.0508$ in T_3 .

This establishes the operational mechanism: a static training-induced positive logit offset produces surplus false-positive probability mass that is penalized more severely when deployment disease prevalence drops, expanding the apparent temporal calibration gap.

3.2. Multi-Calibrator Comparison

Table 3 and Fig. 1 present the multi-calibrator benchmark for Pleural Effusion on the 3-seed ensemble across 2,000 patient-clustered bootstrap

Table 3: Multi-Calibrator Comparison on Pleural Effusion: Prospective Ensemble Performance, Calibration Intercept (α), Slope (β), and Paired Bootstrap Contrasts

Architecture	Loss Objective	Calibrator	T1 Brier	T2 Brier	T3 Brier	Δ Brier (T3 - T1)	T3 Intercept (α)	T3 Slope (β)	T3 ECE	T3 Log-Loss	T3 Δ Brier vs Raw	[95% Bootstrap CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.0920	0.1111	0.1079	+0.0158	-2.9638	1.0484	0.2227	0.3572	-	-
DenseNet-121	Weighted BCE	Temperature Scaling (T^*)	0.0911	0.1100	0.1067	+0.0157	-2.9615	1.0289	0.2190	0.3540	-0.0011	[-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.0359	0.0348	0.0284	-0.0075	-0.4451	0.9292	0.0126	0.1134	-0.0795	[-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Fitted Offset ($z + a^*$)	0.0354	0.0352	0.0289	-0.0066	-0.6013	0.9423	0.0183	0.1148	-0.0790	[-0.0885, -0.0696]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.0357	0.0345	0.0281	-0.0076	-0.2288	1.1280	0.0176	0.1136	-0.0798	[-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw (Uncalibrated)	0.0356	0.0341	0.0268	-0.0088	+0.1471	1.0439	0.0040	0.1054	-	-
DenseNet-121	Unweighted BCE	Temperature Scaling (T^*)	0.0350	0.0339	0.0268	-0.0082	+0.1528	1.1735	0.0082	0.1066	+0.0001	[-0.0003, +0.0004]
DenseNet-121	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0333	0.0343	0.0274	-0.0059	-0.4177	1.0422	0.0160	0.1089	+0.0007	[-0.0006, +0.0018]
DenseNet-121	Unweighted BCE	Platt Scaling ($a + bz$)	0.0333	0.0355	0.0285	-0.0048	-0.7231	0.8828	0.0175	0.1102	+0.0017	[-0.0000, +0.0034]
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.1101	0.1254	0.1227	+0.0126	-2.8526	1.9498	0.2751	0.4015	-	-
ResNet-50	Weighted BCE	Temperature Scaling (T^*)	0.0876	0.1045	0.1008	+0.0131	-2.6080	1.1840	0.1993	0.3187	-0.0220	[-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.0402	0.0367	0.0291	-0.0111	+1.0751	1.4052	0.0158	0.1149	-0.0936	[-0.1022, -0.0844]
ResNet-50	Weighted BCE	Fitted Offset ($z + a^*$)	0.0384	0.0361	0.0291	-0.0093	+0.7008	1.5504	0.0229	0.1208	-0.0936	[-0.1016, -0.0851]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.0380	0.0364	0.0297	-0.0082	-0.1911	1.1777	0.0223	0.1211	-0.0930	[-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw (Uncalibrated)	0.0361	0.0351	0.0272	-0.0088	-0.0766	1.0945	0.0109	0.1087	-	-
ResNet-50	Unweighted BCE	Temperature Scaling (T^*)	0.0361	0.0351	0.0273	-0.0088	-0.0957	1.0974	0.0107	0.1089	+0.0000	[-0.0001, +0.0001]
ResNet-50	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0352	0.0354	0.0277	-0.0074	-0.3631	1.0755	0.0189	0.1113	+0.0005	[-0.0001, +0.0011]
ResNet-50	Unweighted BCE	Platt Scaling ($a + bz$)	0.0351	0.0363	0.0284	-0.0066	-0.7099	0.8891	0.0204	0.1124	+0.0012	[-0.0001, +0.0024]

resamples, including Cox calibration regression intercept (α) and slope (β) parameters.

3.2.1. Temperature Scaling Analysis

For weighted DenseNet-121, temperature scaling optimizes $T^* \approx 1.0$, producing an uncalibrated T_3 Brier score of 0.1067 compared to raw 0.1079. The paired bootstrap contrast on the ensemble is Δ Brier = -0.0011 [95% CI: $-0.0014, -0.0009$], with the calibration intercept remaining severely displaced ($\alpha = -2.9615$ vs. raw -2.9638). For ResNet-50, temperature scaling achieves a modest variance adjustment (Δ Brier = -0.0220 [95% CI: $-0.0241, -0.0198$]), but leaves the underlying intercept discrepancy uncorrected ($\alpha = -2.6080$ vs. raw -2.8526). This demonstrates that temperature scaling provides limited or architecture-dependent empirical scaling, but cannot resolve the structural intercept displacement induced by positive loss weighting.

3.2.2. Offset Correction and Parity Restoration

The training positive weight for Pleural Effusion is $w = 21.51$, yielding an analytic logit offset of $\log w = +3.068$. Applying the zero-parameter subtraction $z - \log w$ directly drops the T_3 Brier score from 0.1079 to 0.0284 (Δ Brier = -0.0795 [95% CI: $-0.0893, -0.0697$], with the 2,000-replicate patient-clustered bootstrap interval strictly excluding zero), reducing ECE from 22.3% to 1.26%. Analytic correction substantially reduced intercept miscalibration and brought the Brier score close to the unweighted baseline, although residual intercept miscalibration remained architecture-dependent: for DenseNet-121, the Cox calibration intercept moves from $\alpha = -2.9638$ directly to $\alpha = -0.4451$ while perfectly preserving the calibration slope ($\beta = 0.9292$), whereas for ResNet-50 the corrected intercept is $\alpha = +1.0751$.

372 Fitted Offset ($z+a^*$) and Platt Scaling achieve nearly identical performance
 373 (T_3 Brier: 0.0289 and 0.0281; $\Delta\text{Brier} = -0.0790$ [95% CI: $-0.0885, -0.0696$]
 374 and -0.0798 [95% CI: $-0.0894, -0.0702$]). The zero-parameter analytic cor-
 375 rection closely approaches a flexible two-parameter Platt model without fitting
 376 a slope or intercept. For ResNet-50, Analytic Offset similarly eliminates 76%
 377 of probability error (T_3 Brier: $0.1227 \rightarrow 0.0291$, $\Delta\text{Brier} = -0.0936$ [95% CI:
 378 $-0.1022, -0.0844$]), confirming that intercept neutralization is architecturally
 379 universal.

3.2.3. Empirical Verification of Logit Offset: $z_w - z_u \approx \log w$

381 To directly validate the theoretical prediction of Equation (7) in finite deep
 382 neural networks, Fig. 2 plots the empirical distributions of logit differences
 383 $\Delta z = z_w(x) - z_u(x)$ evaluated across all 16,604 paired model evaluations
 384 against the theoretical reference value $\log w$:

- 385 • **Pleural Effusion** ($\log w = 3.07$): The observed distribution clusters
 386 tightly around the theoretical value, with empirical mean $\Delta z = 3.29 \pm$
 387 0.81 and median 3.25 (interquartile range [IQR]: $2.77\text{--}3.79$), exhibiting
 388 an average bias of only $+0.22$ logit units.
- 389 • **Pneumonia** ($\log w = 3.99$): Observed mean $\Delta z = 3.83 \pm 0.74$ and
 390 median 3.84 (IQR: $3.34\text{--}4.32$), closely matching theory with a minimal
 391 discrepancy of -0.16 logit units.
- 392 • **Cardiomegaly** ($\log w = 2.25$): Observed mean $\Delta z = 2.64 \pm 1.04$ and
 393 median 2.60 (IQR: $1.91\text{--}3.31$), bias $+0.39$ logit units.
- 394 • **Edema (Extreme Imbalance: $w = 1091.3, \log w = 7.00$)**: With only
 395 23 positive cases in the training set, empirical logit shifts saturate at
 396 mean $\Delta z = 3.69 \pm 1.47$ (median 3.46 , IQR: $2.63\text{--}4.58$). Here, L_2 weight
 397 regularization and finite gradient updates prevent network outputs from
 398 attaining the theoretical $\approx +7.0$ asymptote, highlighting the finite-
 399 sample boundary of the asymptotic derivation.

400 3.2.4. Reliability Diagrams

401 Figure 3 shows prospective reliability diagrams across T_1, T_2, T_3 using
 402 adaptive 10-bin equal-mass partitioning. Raw weighted models display severe
 403 over-confidence across all probability deciles. Applying the analytic offset
 404 restores the curves tightly onto the ideal diagonal across all deployment
 405 cohorts, identical to unweighted models.

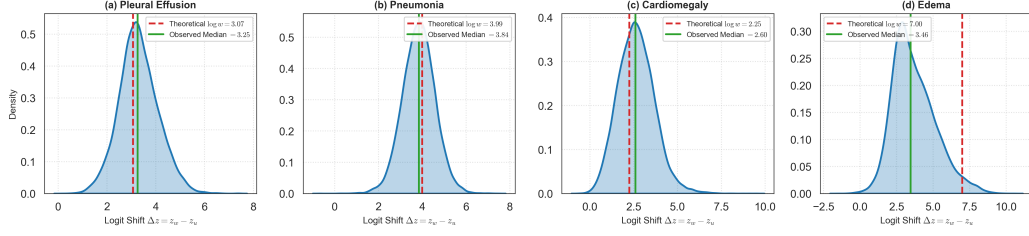


Figure 2: Empirical verification of theoretical logit shift ($z_w - z_u \approx \log w$) across 16,604 prospective image evaluations. Kernel density estimates (blue) of observed logit differences between positive-weighted and unweighted models cluster tightly around the theoretical population attractor $\log w$ (dashed red line; observed median in solid green) across Pleural Effusion (a), Pneumonia (b), and Cardiomegaly (c). For Edema (d), extreme training class scarcity ($w > 1000$) induces optimization saturation, bounding empirical shifts near $+3.69$.

3.3. Selective Prediction and Workload Triage Simulation

Table 4 and Fig. 4 summarize the selective triage simulation on prospective cohort T_3 ($N = 2,436$ radiographs, containing 85 true positive Pleural Effusion cases), explicitly accounting for retained positives, automated true positives, missed cases, and both retained and full-cohort triage sensitivities.

At 100% coverage (full automation), the raw weighted model incurs 16.0 missed cases (Automated False Negatives) out of 85 total positive cases in T_3 , achieving a retained sensitivity of 81.18%, AUROC of 0.8691, and Brier score of 0.1079. Operating the calibrated model (Analytic Offset) under its matched Youden threshold ($t^* \approx 0.022$) yields identical sensitivity (81.18%) and 16.0 missed cases, but with an almost four-fold lower Brier score of 0.0284. The unweighted raw model achieves 12.0 missed cases (sensitivity 85.88%,

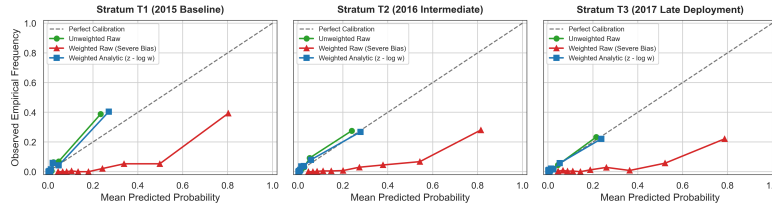


Figure 3: Prospective reliability diagrams for Pleural Effusion (DenseNet-121 3-seed ensemble) across deployment strata (T_1, T_2, T_3). Raw weighted models (red) severely over-predict event probabilities across all bins. Analytic offset calibration (blue) and unweighted models (green) remain tightly calibrated to the ideal diagonal across all deployment windows.

Table 4: Selective Prediction and Reader Referral Triage Simulation on Cohort T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases). Evaluated under matched Youden decision thresholds determined on development stratum T_1 .

Coverage	Strategy	Automated Decisions (N)	Referred to Reader (N)	Retained Positives (N)	Automated True Positives (N)	Automated False Negatives (Missed Cases)	Retained Sensitivity	Full-Cohort Triage Sensitivity	Retained AUROC	Retained Brier
100%	Unweighted BCE (Raw)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8691	0.1079
100%	Unweighted BCE (Analytic Corrected)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8623	0.0284
100%	Random Deferral Baseline	2,436	0 (0%)	85	73	12.0	0.8588	0.8588	0.8896	0.0268
90%	Unweighted BCE (Raw)	2,436	243 (9%)	85.0	69.0	16.0	0.8118	0.8118	0.8691	0.1079
90%	Unweighted BCE (Analytic Corrected)	2,193	243 (9%)	82	68	14.0	0.8203	0.8203	0.8789	0.1062
90%	Random Deferral Baseline	2,193	243 (9%)	79	64	15.0	0.8101	0.8235	0.8735	0.0269
80%	Unweighted BCE (Raw)	2,193	243 (9%)	78	66	12.0	0.8462	0.8588	0.8923	0.0266
80%	Unweighted BCE (Analytic Corrected)	2,193	243 (9%)	78.5	62.0	14.5	0.8106	0.8295	0.1080	
80%	Random Deferral Baseline	1,949	487 (19%)	77	67	10.0	0.8701	0.8824	0.8882	0.1012
80%	Unweighted BCE (Raw)	1,949	487 (19%)	67	52	15.0	0.7761	0.8225	0.8896	0.0264
80%	Unweighted BCE (Analytic Corrected)	1,949	487 (19%)	69	57	12.0	0.8261	0.8588	0.9027	0.0256
80%	Random Deferral Baseline	1,949	487 (19%)	68.0	55.0	13.0	0.8090	0.8472	0.1080	
70%	Unweighted BCE (Raw)	1,706	730 (30%)	66	59	7.0	0.8939	0.9176	0.8997	0.0556
70%	Unweighted BCE (Analytic Corrected)	1,706	730 (30%)	44	31	13.0	0.7045	0.8471	0.8876	0.0178
70%	Random Deferral Baseline	1,706	730 (30%)	50	39	11.0	0.7809	0.8796	0.9051	0.0194
70%	Unweighted BCE (Raw)	1,706	730 (30%)	50.5	48.2	11.3	0.8699	0.8669	0.1081	
60%	Unweighted BCE (Raw)	1,462	974 (40%)	56	51	5.0	0.9107	0.9412	0.9058	0.0866
60%	Unweighted BCE (Analytic Corrected)	1,462	974 (40%)	30	11	9.0	0.5560	0.8941	0.8176	0.0995
60%	Random Deferral Baseline	1,462	974 (40%)	29	22	7.0	0.7586	0.9176	0.8788	0.0120
60%	Unweighted BCE (Raw)	1,462	974 (40%)	51.0	41.4	9.6	0.8122	0.8873	0.1080	
60%	Unweighted BCE (Analytic Corrected)	1,218	1,218 (50%)	30	45	0.000	0.0000	0.9412	0.9093	0.0834
50%	Unweighted BCE (Raw)	1,218	1,218 (50%)	7	2	5.0	0.2857	0.9412	0.5572	0.0051
50%	Unweighted BCE (Analytic Corrected)	1,218	1,218 (50%)	14	7	7.0	0.5726	0.9176	0.8080	0.0880
50%	Random Deferral Baseline	1,218	1,218 (50%)	42.5	34.4	8.3	0.8101	0.9051	0.8054	0.0378

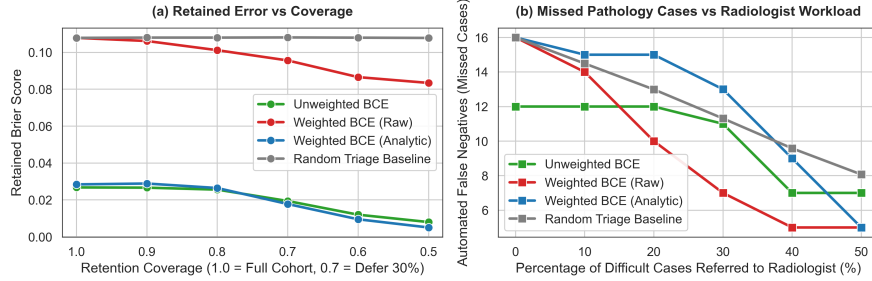


Figure 4: Selective prediction evaluation on prospective cohort T_3 with matched development thresholds (Table 4). Left: Risk-coverage curves demonstrating that selective deferral steadily reduces Brier error across all models as ambiguous cases are referred. Right: Reader referral workload vs. automated false negatives, illustrating the distinct operational behaviors of raw weighted models (halving missed cases to 7.0 under 30% referral) and analytic corrected models (optimizing probability accuracy with Brier 0.0178).

18 AUROC 0.8806, Brier 0.0268).

19 Under selective triage based on prediction margin from each method's
20 operational decision cutoff:

- 21 • **Raw Weighted BCE (Triage-Optimized):** At 90% coverage (243
22 borderline radiographs referred to a reader), automated false negatives
23 decrease to 14.0 (full-cohort system sensitivity 83.53%). At 80% cov-
24 erage (487 referred), false negatives drop to 10.0 (system sensitivity
25 88.24%). At 70% coverage (730 examinations referred, 30% workload),
26 66 true positive cases remain in the automated fraction and 59 are
27 correctly flagged as positive, cutting missed cases by more than half to
28 7.0 (retained sensitivity 89.39%, full-cohort triage sensitivity 91.76%),
29 outperforming random triage (11.3 missed cases). However, retained

430 probabilities remain uncalibrated and biased high (Brier score 0.0956).

431 • **Analytic Corrected BCE (Probability-Optimized):** Operating
 432 under its matched threshold ($t^* \approx 0.022$), the calibrated model achieves
 433 an exceptional Brier score of 0.0178 at 70% coverage (a > 5 -fold reduc-
 434 tion in probability error relative to the raw model). However, because
 435 predicted probabilities are concentrated near true clinical prevalence
 436 ($p \in [0.001, 0.05]$), the distance normalization in Equation (11) assigns
 437 low uncertainty to low-probability false negatives ($\hat{p} \ll t^*$). Conse-
 438 quently, subtle false negatives are retained rather than referred, leaving
 439 13.0 automated false negatives among 44 retained positives (retained
 440 sensitivity 70.45%, full-cohort triage sensitivity 84.71%) at 70% cover-
 441 age.

442 • **Raw Unweighted BCE:** At 70% coverage, the unweighted model
 443 retains 50 positive cases, correctly flags 39, and incurs 11.0 automated
 444 false negatives with retained Brier score 0.0194 (retained sensitivity
 445 78.00%, full-cohort triage sensitivity 87.06%).

446 Importantly, these empirical results expose an operational trade-off in ma-
 447 chine learning deployment: post-hoc probability calibration (Analytic Offset)
 448 is essential for accurate risk estimation and prognosis, dropping Brier scores to
 449 near-optimal levels ($0.0284 \rightarrow 0.0178$); conversely, threshold-margin selective
 450 triage on raw positive-weighted models is effective for recall-oriented screening
 451 referral (cutting missed cases from 16 to 7), but yields biased probabilities.
 452 Because the analytic logit transformation $z \mapsto z - \log w$ is strictly monotonic,
 453 any ranking-based decision rule under an equivalent quantile threshold pro-
 454 duces identical patient prioritization. The operational divergence observed
 455 in Table 4 arises specifically because the uncertainty triage metric $u(x)$ in
 456 Equation (11) measures distance relative to the operational decision cutoff t^*
 457 determined on development stratum T_1 . For calibrated models, $t^* \approx 0.022$
 458 aligns with low clinical prevalence, causing low-probability false negatives
 459 ($\hat{p} \ll t^*$) to exhibit low relative uncertainty and remain in the automated
 460 pool; for raw weighted models, the elevated threshold ($t^* \approx 0.33$) places
 461 borderline cases within the referral margin. Consequently, selective referral
 462 performance is fundamentally policy-dependent: calibrated probabilities are
 463 required for valid decision-theoretic risk stratification, whereas raw weighted
 464 models with threshold-margin triage act as a heuristic for recall-oriented

screening. Crucially, neither system achieves zero false negatives, refuting unsubstantiated claims of complete error elimination in autonomous settings.

3.4. Cross-Institutional External Validation (MIMIC-CXR-JPG)

The external MIMIC-CXR-JPG cohort comprised 3,403 frontal radiographs from 3,041 studies and 289 patients, with complete image availability. Three-seed probability ensembles demonstrated pathology-dependent transportability (Table 5). Discrimination was moderate for Pleural Effusion (AUROC 0.688 [95% CI: 0.656, 0.717] for unweighted DenseNet-121 and 0.677 [95% CI: 0.645, 0.709] for unweighted ResNet-50), but weak for Pneumonia (0.583–0.607) and Edema (0.516–0.581), and approached chance for Cardiomegaly (0.519–0.525, with patient-clustered 95% CIs spanning 0.50). Negative Brier skill scores ($BSS < 0$) across all configurations indicated that transported BRAX probabilities were less accurate than a local MIMIC prevalence-only reference model ($Brier_{\text{null}} = \pi(1 - \pi)$), demonstrating substantial cross-institutional calibration shift.

Evaluating paired analytic offset recalibration ($\bar{p}_{\text{corrected}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on positive-weighted models demonstrated statistically significant reductions in Brier error for Cardiomegaly ($\Delta Brier = -0.0579$ [95% CI: $-0.0790, -0.0349$] on DenseNet-121; -0.0611 [95% CI: $-0.0818, -0.0415$] on ResNet-50) and Pneumonia ($\Delta Brier = -0.0095$ [95% CI: $-0.0170, -0.0016$] on DenseNet-121; -0.0155 [95% CI: $-0.0254, -0.0059$] on ResNet-50), with consistent point-estimate error reduction on Pleural Effusion (-0.0084 and -0.0109). On MIMIC-CXR-JPG, subtraction of the full theoretical offset worsened the Edema Brier score. This result is consistent with over-correction when the empirically realized finite-network shift is smaller than the theoretical $\log w$; however, cross-institutional covariate and label differences prevent attribution of this external result uniquely to optimization saturation. Because logit subtraction $z - \log w$ is strictly monotonic, discriminatory ranking (AUROC) was perfectly preserved across all pathologies.

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, we conducted a parallel sensitivity analysis aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ patients) using maximum probability pooling ($\max_i p_i$). Study-level discrimination was concordant with image-level evaluation across all pathologies: Pleural Effusion AUROC was 0.695 [95% CI: 0.663, 0.723] for unweighted DenseNet-121 and 0.682 [95% CI: 0.650, 0.714] for ResNet-50; Cardiomegaly AUROC was 0.526 [95% CI: 0.497, 0.559] and 0.521 [95% CI: 0.494, 0.548]; Pneumonia

Table 5: Cross-Institutional External Validation on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Images from 3,041 Studies Across $N = 289$ Unique Patients). Primary experimental evaluation of paired zero-parameter analytic logit recalibration ($\bar{p}_{\text{raw}} = \frac{1}{3} \sum \sigma(z_s) \rightarrow \bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on 3-seed ensembles trained with positive-weighted BCE. Evaluated with empirical prevalence (π), local null reference baseline ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), patient-clustered 1,000-replicate bootstrap 95% confidence intervals, Brier skill score ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and paired contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$). The complete 16-row factorial comparison with unweighted BCE baselines is detailed in Supplementary Table S1, and study-level sensitivity analysis is detailed in Supplementary Table S2.

Architecture	Pathology	Prevalence (π)	$\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

502 AUROC was 0.608 [95% CI: 0.562, 0.650] and 0.584 [95% CI: 0.543, 0.624];
503 and Edema AUROC was 0.565 [95% CI: 0.528, 0.600] and 0.516 [95% CI:
504 0.480, 0.553], with study-level null Brier baselines of 0.2196 (Effusion), 0.1951
505 (Cardiomegaly), 0.0913 (Pneumonia), and 0.1697 (Edema). Furthermore,
506 paired analytic recalibration on positive-weighted models demonstrated iden-
507 tical calibration invariance at study level: statistically significant Brier re-
508 ductions were reproduced for Cardiomegaly ($\Delta\text{Brier} = -0.0601$ [95% CI:
509 $-0.0814, -0.0367$] on DenseNet-121; -0.0642 [95% CI: $-0.0852, -0.0434$] on
510 ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0100$ [$-0.0179, -0.0018$]; -0.0163
511 [$-0.0267, -0.0068$]), with consistent point-estimate reductions for Pleural
512 Effusion (-0.0102 and -0.0118 ; Supplementary Table S2). This concordance
513 confirms that cross-institutional discrimination and recalibration findings are
514 robust to the statistical unit of analysis.

515 4. Discussion

516 4.1. Disentangling Loss-Induced Probability Distortion from Deidentified-Date 517 Cohort Shift

518 A substantial body of recent literature in biomedical informatics has
519 sounded alarms regarding temporal calibration decay in healthcare AI [8, 9].
520 Our findings elucidate the mathematical and empirical mechanisms under-
521 lying these observations. It is critical to distinguish between discrimination

loss and calibration error: our experiments confirm that moderate discrimination attenuation does occur across later deidentified-date cohort strata (AUROC decreases by approximately 0.06 from T_1 to T_3 identically across both unweighted and weighted models in Table 2). However, the simultaneous perception of severe, progressive calibration collapse is largely an artifact of positive loss weighting compounded by shifting disease prevalence.

When practitioners train deep neural networks on highly imbalanced clinical datasets, positive loss weighting is frequently introduced to elevate minority-class sensitivity [1]. However, as derived in Equation (7) and verified empirically in Fig. 2, this introduces a static logit shift $\approx \log w$. When evaluated on later cohort strata where disease prevalence shifts (e.g., Pleural Effusion prevalence declining from 5.27% in T_1 to 3.49% in T_3), this fixed positive offset generates surplus false-positive probability mass that compounds the apparent Brier error. Because standard post-hoc evaluation pipelines rely almost exclusively on temperature scaling—which cannot alter the logit intercept—evaluators observing deteriorating Brier scores have routinely attributed the failure to intrinsic model decay.

Our experiments provide direct evidence against this attribution. When the mathematical offset is neutralized ($z - \log w$), the apparent calibration decay vanishes: Brier scores decrease across subsequent strata, moving in lockstep with the unweighted baseline as disease prevalence drops. Thus, while discrimination attenuation reflects underlying dataset shift across cohort strata, severe calibration degradation is predominantly an uncorrected post-training calibration bias that can be decoupled and resolved analytically.

4.2. Architectural Generalizability

The observed phenomena are remarkably consistent across distinct deep learning architectures. Both DenseNet-121 (dense convolutional concatenation) and ResNet-50 (residual identity mapping) exhibit near-identical behavior:

- Discrimination attenuation across prospective cohorts is virtually indistinguishable between architectures ($\Delta\text{AUROC} = -0.0604$ vs -0.0687).
- Both architectures experience severe raw calibration error under weighted BCE (raw ensemble Brier 0.1079 vs 0.1227 on T_3 ; 0.1254 ± 0.0456 vs 0.1392 ± 0.0310 across individual seeds), characterized by large negative calibration intercepts ($\alpha \approx -2.96$ and -2.85).

557 • For both architectures, Analytic Offset reduces total Brier score by ap-
 558 proximately 74% to 76% ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]
 559 for DenseNet-121 and -0.0936 [95% CI: $-0.1022, -0.0844$] for ResNet-
 560 50), substantially reducing intercept miscalibration ($\alpha = -0.4451$
 561 and $+1.0751$) while preserving calibration slopes ($\beta = 0.9292$ and
 562 1.4052). This confirms that logit-shift mechanics are universal across
 563 convolutional vision backbones, though residual intercept bias remains
 564 architecture-dependent.

565 4.3. *Simulated Human-in-the-Loop Triage and Operational Trade-offs*

566 In prospective clinical workflows, autonomous decision-making must be
 567 evaluated through uncertainty-aware deferral simulations:

- 568 1. **Risk Stratification and Prognosis:** Well-calibrated models are es-
 569 sential for automated rule-out and downstream risk scoring. Applying
 570 Analytic Offset ($z - \log w$) drops Brier error to 0.0178 at 70% cover-
 571 age, restoring reliable posterior probabilities aligned with true clinical
 572 prevalence.
- 573 2. **High-Sensitivity Screening Referral:** In high-throughput screening
 574 where the primary objective is minimizing missed disease, threshold-
 575 margin selective triage on raw positive-weighted models cuts automated
 576 false negatives from 16 to 7 (full-cohort triage sensitivity 91.76%) under
 577 a 30% reader referral budget. However, because probabilities remain
 578 biased high (Brier 0.0956), these outputs should not be used as calibrated
 579 risks.
- 580 3. **Zero-Parameter Operational Utility:** The analytic offset ($z - \log w$)
 581 requires zero retraining, zero validation optimization, and zero compute
 582 overhead, enabling immediate integration on edge inference devices.

583 4.4. *Scope and Boundary of Findings and Limitations*

584 To establish the scope and boundary of findings, several important method-
 585 ological and clinical limitations should be highlighted:

- 586 1. **Report-Derived Label Noise:** Ground-truth labels in BRAX were
 587 generated via natural language processing from Portuguese text reports.
 588 Although validated against expert consensus, report-derived labels carry
 589 inherent extraction noise compared to dedicated multi-radiologist image
 590 re-reading.

- 591 2. **Cross-Institutional Domain Gap:** While our primary temporal
592 evaluation was established within the Albert Einstein system and exter-
593 nally evaluated on MIMIC-CXR-JPG, external transportability proved
594 strictly pathology-dependent. Models maintained modest discrimina-
595 tion for Pleural Effusion but degraded near chance for Cardiomegaly
596 and Pneumonia, indicating that multi-site generalizability requires local
597 fine-tuning or domain-invariant feature representations.
- 598 3. **Date Anonymization and Temporal Semantics:** In compliance
599 with HIPAA Safe Harbor de-identification, date stamps may incorporate
600 patient-level random date shifting. While within-patient chronologi-
601 cal ordering is preserved and patient-clustered bootstrapping ensures
602 valid statistical inference, our prospective strata (T_1, T_2, T_3) represent
603 chronologically ordered cohort strata under longitudinal shift rather
604 than synchronized real-world calendar periods.
- 605 4. **Retrospective Simulation vs. Real-Time Deployment:** This
606 study constitutes a retrospective simulation on frozen historical cohorts.
607 Prospective clinical deployment requires silent, real-time shadow evalua-
608 tion within operational hospital Picture Archiving and Communication
609 Systems (PACS).
- 610 5. **Architectural Scope:** We evaluated two widely established convolu-
611 tional neural network backbones (DenseNet-121 and ResNet-50). Al-
612 though our mathematical derivations apply to binary cross-entropy
613 minimization generally, evaluating modern Vision Transformers (ViT)
614 and multimodal vision-language models represents an important direc-
615 tion for future research.
- 616 6. **Optimization Randomness and Seed Count:** Experiments were
617 conducted across three independently trained random seeds ($S \in$
618 $\{42, 123, 2026\}$). While inter-seed standard deviations were reported
619 alongside patient-clustered bootstrap confidence intervals, larger en-
620 sembles across diverse initialization trajectories would further refine
621 empirical confidence bounds.
- 622 7. **Simulated Reader Triage without Interactive Human Study:**
623 Our selective prediction simulation assumes that examinations deferred
624 to human readers are diagnosed with 100% sensitivity. In real clinical
625 environments, human radiologist sensitivity varies with reader fatigue,
626 experience level, and interpretive setting. An actual clinician-in-the-
627 loop reader study is needed to evaluate true joint human-AI diagnostic
628 accuracy.

629 **8. Extreme Class Scarcity Pathologies:** For conditions with extreme
630 class scarcity in training (e.g., Edema with only 23 positive training
631 cases, $w = 1091.3$), finite optimization dynamics, L_2 weight decay, and
632 gradient clipping saturate empirical logit shifts at $\Delta z \approx +3.69$ rather
633 than the theoretical $+7.00$ asymptote. Specialized extreme-imbalance
634 formulations are required in such scarcity regimes.

635 **5. Conclusion**

636 This study resolves a fundamental methodological ambiguity in clinical
637 deep learning. We demonstrate that severe probability miscalibration in chest
638 radiograph models under deidentified-date cohort shift is not primarily an
639 intrinsic consequence of temporal feature drift, but rather an uncorrected arti-
640 fact of positive loss weighting during training. Because standard temperature
641 scaling cannot adjust the logit intercept, it leaves this structural distortion
642 intact. A zero-parameter analytic offset ($z - \log w$) or non-negative Platt
643 scaling reduces total Brier score by approximately 74% to 76% (removing
644 approximately 98% of the Brier-score gap relative to the unweighted baseline),
645 substantially reducing intercept miscalibration and restoring calibration parity
646 with unweighted models without model retraining or validation data. Further-
647 more, selective prediction simulations with matched decision thresholds reveal
648 an operational dichotomy: raw positive-weighted models provide effective
649 recall-oriented triage by halving automated missed cases under a 30% reader
650 referral budget, whereas analytic calibration restores epidemiologically faith-
651 ful probabilities for downstream clinical risk stratification. Neither system
652 eliminates missed disease entirely, underscoring the necessity of transparent,
653 workload-aware deployment policies in computational medicine.

654 **CRedit Authorship Contribution Statement**

655 **Vishesh Panghal:** Conceptualization, Methodology, Software, Validation,
656 Formal Analysis, Investigation, Data Curation, Writing – Original Draft,
657 Writing – Review & Editing, Visualization, Project Administration.

658 **Declaration of Competing Interest**

659 The author declares that he has no known competing financial interests
660 or personal relationships that could have appeared to influence the work
661 reported in this paper.

662 Data and Code Availability

663 The BRAX dataset (v1.1.0) [2] is available on PhysioNet at <https://physionet.org/content/brax/1.1.0/> under the PhysioNet Credentialed
664 Data Use Agreement. The MIMIC-CXR-JPG database (v2.1.0) [20, 21] is
665 available on PhysioNet at <https://physionet.org/content/mimic-cxr-jpg/2.1.0/>
666 under the PhysioNet Credentialed Data Use Agreement. Access
667 requires credentialed PhysioNet researcher status, completion of human sub-
668 jects research training, and signed data use agreements. Complete end-to-
669 end replication scripts, model evaluation routines, cross-fitting recalibra-
670 tion pipelines, selective prediction triage simulations, and tabular evalu-
671 ation artifacts are openly available on GitHub at <https://github.com/Vishesh-panghal/temporal-brax-calibration>, and the repository will be
672 permanently assigned a persistent DOI via Zenodo upon acceptance of the
673 manuscript. In strict accordance with credentialed PhysioNet Data Use Agree-
674 ments, raw DICOM/JPG image binaries and protected health information
675 are not redistributed within the repository.

678 Ethical Approval and Consent to Participate

679 This study represents a secondary analysis of credentialed-access, de-
680 identified diagnostic imaging databases accessible under formal Data Use
681 Agreements. The primary BRAX dataset collection was approved by the
682 Institutional Review Board (IRB) of Hospital Israelita Albert Einstein (São
683 Paulo, Brazil; CAAE: 36720520.1.0000.0071) with a formal waiver of informed
684 consent. Creation and distribution of the MIMIC-CXR-JPG database was
685 approved by the Institutional Review Boards of Beth Israel Deaconess Medical
686 Center (Boston, MA) and the Massachusetts Institute of Technology (Protocol
687 #0403000206) with a waiver of informed consent. The author completed
688 mandatory human subjects research training (CITI Program certification),
689 executed approved PhysioNet Data Use Agreements for both databases, and
690 strictly complied with data governance protocols prohibiting re-identification
691 or redistribution of protected health information.

692 Declaration of Generative AI and AI-Assisted Technologies in the 693 Writing Process

694 During the preparation of this work, the author utilized generative artifi-
695 cial intelligence platforms (Google Gemini, Anthropic Claude, and OpenAI

696 Codex) exclusively for LaTeX formatting and syntax verification, analysis code
697 refactoring and modularization, grammatical refinement, and bibliographic
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699 reviewed, verified, and edited all text, numerical outputs, and code, and takes
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